BACHELOR'S DEGREE DEFENSE

Multi-Hop Biomedical Reasoning

Retrieval-Augmented Generation for Drug Interaction QA Using Local LLMs

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Contents

Problem Statement

The Task

Given a **query drug** and a **candidate list**, identify which candidate interacts through **multi-hop reasoning** across PubMed abstracts and DrugBank.

The Challenge

A single document rarely contains complete evidence. Must **chain facts** through intermediate **bridge entities** across multiple sources.

Constraint

Fully local inference. **No paid APIs**, no fine-tuning. RTX 5050, 8 GB VRAM, 4-bit quantized LLMs (7B-9B params).

INPUT

Query Drug + Candidate List

Multi-Hop QA System

Retrieval + LLM Reasoning

OUTPUT

Correct DrugBank ID

External: PubMed + DrugBank

Dataset & Challenges

342
Dev Questions

1,620
Train Questions

~8.6
Avg Candidates/Q

133
Unique Answer Drugs

EM
Exact Match

Key Challenges

- Multi-hop reasoning requires chaining 2-3 documents
- Bridge entities not explicitly mentioned in query
- Large candidate lists increase search space
- Small models (7-9B) have limited reasoning capacity
- Zero-shot inference only, no fine-tuning allowed

Random Baseline 11.63%

Oracle Upper Bound 76.9%

Evaluation: prediction must exactly match the correct DrugBank ID

Literature Review — Published Systems

System	Model	Params	EM%	Setup	Key Limitation
DMIS Lab (2025)	GPT-4o + GPT-o3	~1.8T (API)	87.3	3-Arm Ensemble + Web Search	High API cost
UETQuintet (2025)	GPT-4o-mini + o3-mini	~1T (API)	83.8	Hop-logic classifier + API	API dependency
NHSRAG (2025)	MedReason-8B + Gemini	8B + API	73.4	NER + sparse search	$451/run fallback cost
CLaC (2025)	Qwen2.5-32B	32B (local)	67.6	Wikipedia retriever + vLLM	40-60hr inference time
Orekhovich (2025)	llama3-med42-8B + Qwen-32B	8B+32B	43.9	Hybrid BM25S + MedEmbed	Sub-optimal on complex hops
lasigeBioTM (2025)	Mistral-7B + MedGemma-27B	7B+27B	28.3	NER + Mondo Ontology	Limited entity coverage
BiDAF (Welbl 2018)	LSTM	~50M	47.8	Extractive RC	Outdated architecture
DeepRAG (2025)	DeepSeek-R1 (671B)	671B	20.7	Hierarchical + DPO	Massive resource needs
CaresAI (2025)	LLaMA-3 8B (LoRA)	8B fine-tuned	18.6	SFT on BioASQ+MedQuAD	Format failures

● Local models ● API-dependent

Research Approach — Six Development Stages

1	Baseline 1: Direct LLM QA BioMistral-7B — parametric knowledge only, no retrieval	15.8%
2	Baselines 2–3: Sparse & Dense RAG BM25 (sparse) + MedCPT (dense) retrieval with BioMistral-7B	16.4%
3	**Stage 3: Hybrid + Qwen3.5-9B** Hybrid Scored Retrieval (BM25+MedCPT) + Guided Prompt at k=3	33.3%
4	**Stage 4: Entity Bridging** LLM extracts bridge drug → bridge-boosted retrieval → answer	33.3%
5	Stage 5: Ensemble — Majority Vote 3-way vote: Stage 3 + Entity Bridge + Ontology Verification	35.4%
6	**Stage 6: Meta-Classifier ★ BEST** Logistic Regression on 1,976 features — selects best path per question	51.17%

Baseline Pipeline — Stage 1: Direct LLM QA

INPUT

Query + Candidate List

[DB00001, DB00002, ...]

PROCESS

BioMistral-7B

No documents retrieved.

Parametric knowledge only.

OUTPUT

Predicted DrugBank ID

15.8%

Exact Match

54 / 342 correct

Why it fails:

Multi-hop drug interactions require chaining evidence across multiple documents. Parametric knowledge alone is insufficient (15.8% EM vs 76.9% oracle).

Entity Bridging Pipeline — Stage 4

STEP 1

Bridge Extraction

- CoT Enriched guided prompt
- Qwen3.5-9B identifies intermediate bridge drug
- No documents retrieved at this stage
- Example: Drug A > Bridge > Drug B

STEP 2

Bridge-Boosted Retrieval

- Hybrid: MedCPT + BM25 retrieval
- Bridge drug name injected as boost signal
- Score = 0.5xMedCPT + 0.5xTermCount + wxBridge
- Top k=3 documents selected (optimal)

STEP 3

Answer Generation

- Qwen3.5-9B selects from candidate list
- Incorporates retrieved docs + bridge hint
- Outputs DrugBank ID as final answer

33.3% EM

114 / 342 correct | +17.5pp over baseline

Key Insight

The bridge drug acts as a **semantic anchor** guiding retrieval toward evidence chain documents.

Ensemble — Majority Vote (Stage 5)

INPUT — Query Drug + Candidate List

Path 1

Stage 3 (Hybrid)

Hybrid Retrieval
+ Qwen3.5-9B
(Guided Prompt)

33.3% individual EM

Path 2

Stage 4 (Entity Bridge)

Bridge-Boosted
Retrieval
+ Qwen3.5-9B

33.3% individual EM

Path 3

Stage 4 (OV-EXP4)

Ontology
Verification
+ Qwen3.5-9B

32.2% individual EM

Majority Vote, Each question: the candidate predicted by ≥2 paths wins; if no majority, Path 2 (best individual) decides

35.4%
(121/342)

Limitation: Blind Voting

Majority vote treats all paths equally. But Path A might be better for certain drug types, while Path B excels on others. A smarter approach would consider **confidence signals** and **drug-specific patterns** to weigh paths differently per question. This motivates the meta-classifier in Stage 6.

Best System — Meta-Classifier Pipeline (51.17% EM)

PATH A
BM25 Sparse
BioMistral-7B, k=3
16.96%

PATH B
MedCPT Dense
Qwen2.5-7B, k=3
19.59%

PATH C
Hybrid RAG
Qwen3.5-9B, k=3
16.08%

Meta-Classifier: Logistic Regression (L1, C=0.01)

Feature Set S4: **14 rank** + **1,962 drug identity** = 1,976 features

Train	Test	AUC	Gap
1,620 Q	342 Q	0.8054	0.036

51.17%

Exact Match

+15.8pp over voting

1. Learns **which path to trust** per question type
2. Uses **confidence signals** and drug identity patterns
3. Drug interactions are **highly drug-pair specific**

Meta-Classifier — Feature Engineering Design

Rank Features (14 — 0.7%)

- Rank positions (1-9) of candidate per path (A/B/C)
- Recall@K scores per path

Drug Identity (1,962 — 99.3%)

- One-hot encoding of candidate DrugBank ID
- 13 of 15 top features are drug identity

Key: Learns which DRUG is correct, not which path is better.

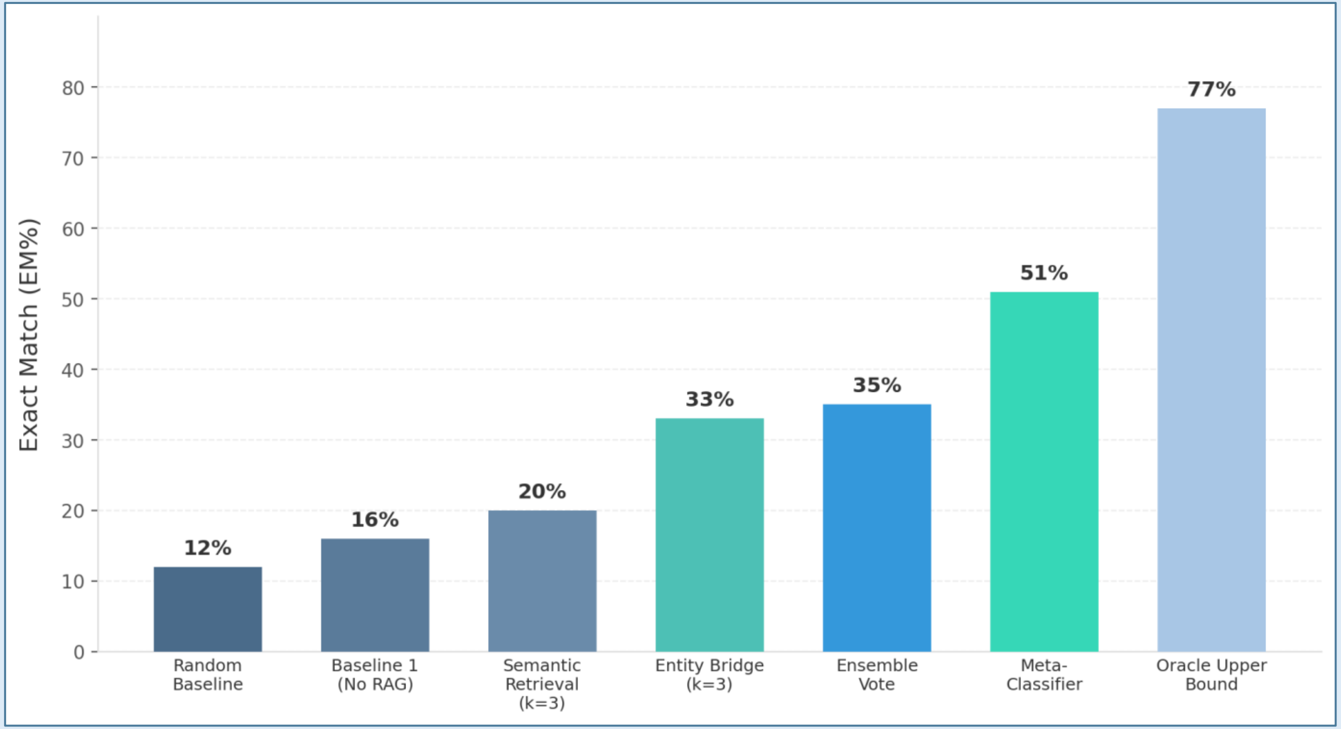
Classifier Design

- Logistic Regression + L1 (C=0.01)
- No leakage: train/train.json, test/dev.json
- AUC 0.8054 | Gap 0.036
- L1 sparsity: only predictive drugs active

Training Data

- 1,620 questions: run 3 paths, collect ranks
- Label = 1 if matches gold DrugBank ID
- Feature: 14 rank + 1,962 binary drug per candidate

Experiments & Results — Stage-by-Stage Progression



15.8% Direct LLM
Parametric knowledge alone

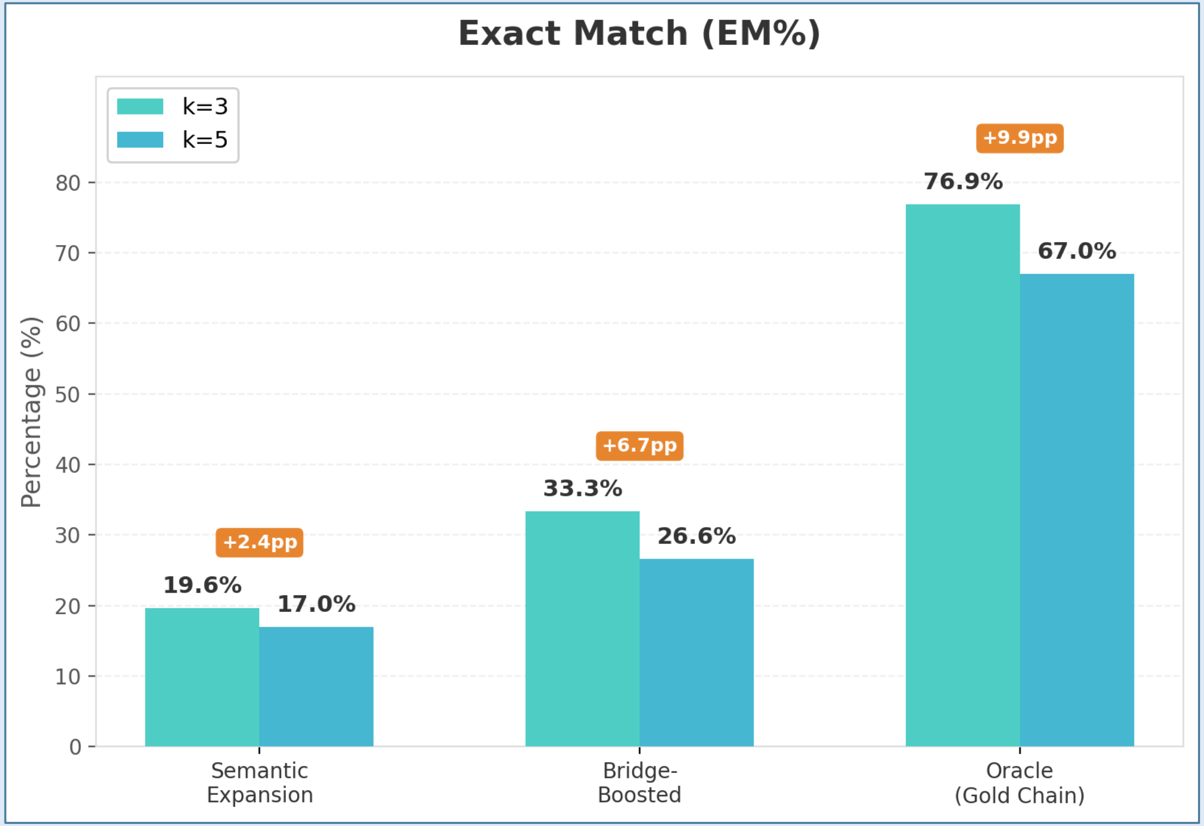
33.3% Entity Bridge
+17.5pp over baseline

35.4% Ensemble
+2.1pp from 3 paths

51.17% Meta-Classifier
+15.8pp over ensemble

76.9% Oracle
Gold chain k=3 upper bound

Retrieval Depth Analysis — Why k=3 Is Optimal



k=3 = 33.3%

Best Result @ k=3

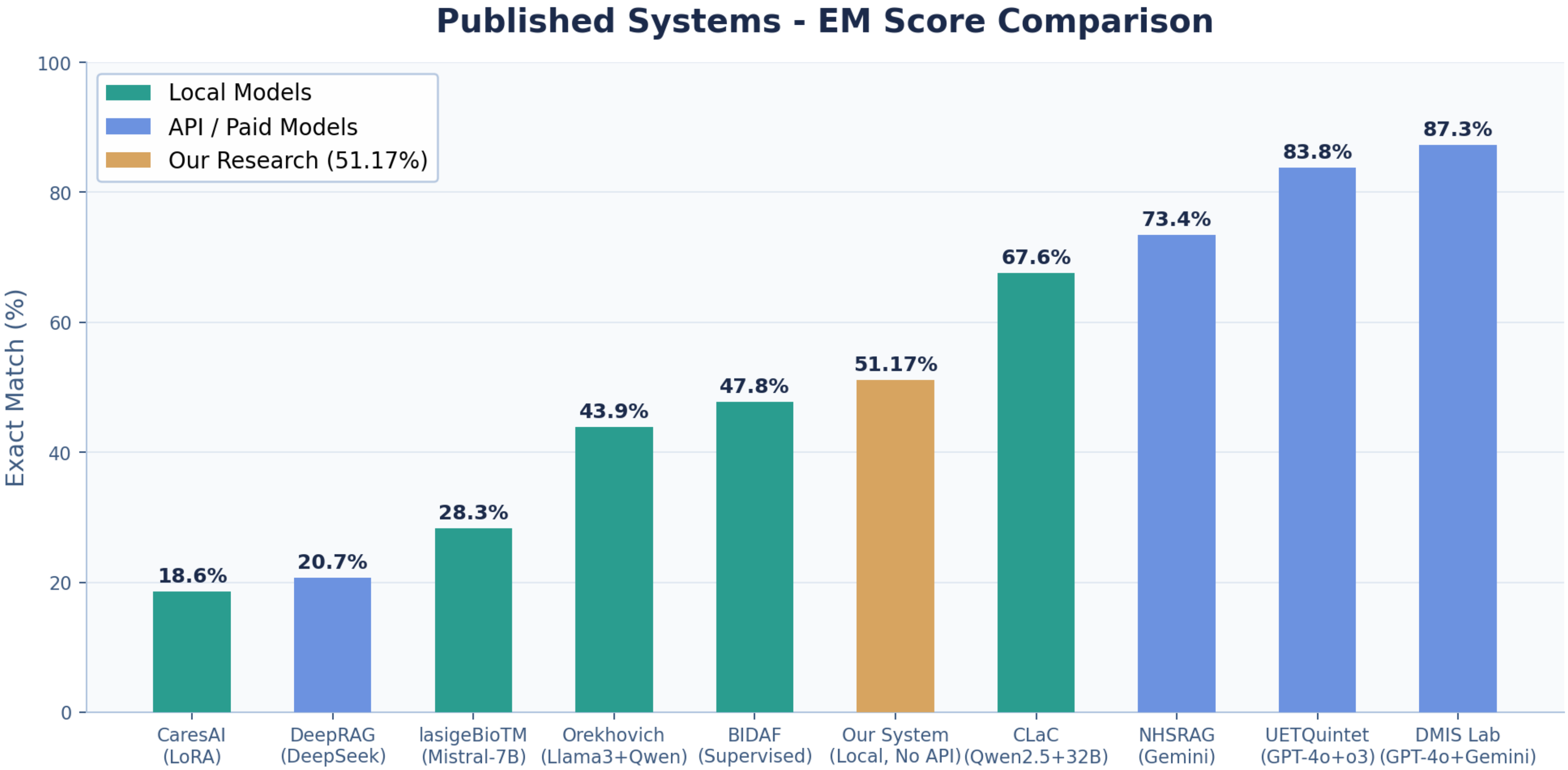
k=3 consistently outperforms k=5

Even Oracle (gold chain) performs worse at k=5 (66.96%) vs k=3 (76.90%). More context adds noise.

Small models, short context

7-9B models struggle to extract answers from long concatenated documents. k=3 wins for ALL methods.

Published Systems — EM Score Comparison



Key Findings & Contributions

1. Model Over Architecture

With BioMistral-7B, adding retrieval hurt. Switching to Qwen3.5-9B alone gained +16.9pp. The model was the bottleneck, not the pipeline.

2. k=3 Is Optimal

Top-3 docs consistently beat top-5 and top-10. Extra context adds noise for small local models.

3. Simplicity Wins

Complex modules (Query Decomposition, Ontology) do not surpass well-tuned simple retrieval at 7-9B scale.

4. Meta-Learning Power

Meta-classifier (51.17%) outperforms voting (35.4%) by +15.8pp. Trusting the right path per question is far more effective than blind aggregation.

5. Drug Identity Dominates

13 of 15 top features are drug-specific embeddings. The classifier learns which drug is correct, not just where it ranks.

Surpasses 4 Published BioCreative IX Systems

Without any fine-tuning, on consumer hardware only

Conclusion & Future Work

Conclusion

- Best result: **51.17% EM** via Meta-Classifier on consumer hardware (RTX 5050, 8 GB VRAM)
- Surpasses 4 published BioCreative IX systems without fine-tuning
- Meta-learning (+15.8pp over voting) proves complementary paths matter
- Retrieval quality is the primary bottleneck: **25.73pp gap** to oracle remains

Future Work

- Improve retrieval to close the 25.73pp gap to oracle
- Explore larger quantized models (70B Q4) for reasoning
- Graph-based reasoning over drug interaction networks
- Apply meta-classifier to other multi-hop QA benchmarks

Thank You

Questions & Discussion

GitHub: Mariam6600/Multi-Hop-Biomedical-Reasoning

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